

European service-hub expansion

Board decision brief • 2025 shipped base • returns known by 31 January 2026

Recommendation

Approve a staged Czechia + Spain program with a maximum €225,000 year-zero capex and five FTE. Open Spain first; authorize Czechia procurement only after the 90-day demand, service, returns and FX gate. This pair has the highest modeled base annual increment among feasible choices, €198,099 after recurring fixed costs, with simple payback of 1.14 years once steady-state benefits are realized. [S1–S5]

The modeled joint stress cuts that increment to €32,180 and stretches payback to 6.99 years. Netherlands + Spain sacrifices €31,168 of base annual increment and requires €45,000 more capex and one extra FTE, but preserves €107,630 under the same stress. If the board requires stress payback below three years, choose Netherlands + Spain instead. The three-year threshold is a proposed risk preference, not a client rule. [S4–S5]

Decision frame

Hard constraint	Status	Basis
At most two hubs	Met: two	Client scenario policy [S5]
Capex ≤ €450,000	Met: €225,000	Client scenario policy and option costs [S4–S5]
Staff ≤ seven FTE	Met: five	Client scenario policy and option staffing [S4–S5]

Choice criteria are modeled annual contribution, year-zero capital, staffing, payback and resilience under the defined joint stress. The first four are computed from the synthetic client inputs; resilience is the same model under a prescribed, non-probabilistic stress. No weights or causal demand estimates are asserted. [S4–S5]

What the evidence supports

The eligible shipped base totals 1,254 orders and 77,436 units. Spain and Czechia have the two largest 2025 contributions, €421,900 and €397,335. This is a client-extract accounting observation, not proof that local hubs cause demand growth. Population and GDP per capita supply context only. [S1–S3, S6–S7]

Conclusion card: Fund the capped, gated Czechia + Spain program. Grounds: the reconciled client base and scenario arithmetic are computed; hub savings/uplift and fixed costs are synthetic assumptions; ECB/World Bank observations are archived official snapshots. Remaining uncertainty: local hub uplift, implementation timing, actual FX/cash rates and return behavior are unverified. Flip: select Netherlands + Spain if the board adopts a ≤3-year stressed payback rule or if pilot evidence lowers the Czechia base increment by more than the €31,168 pair advantage. Defer the second site if its gate fails. Alternative: Netherlands + Spain is stronger under the joint stress but costs more capital and staff. [S1–S7]

1 | Reconciled 2025 economics

All amounts below are euro. Shipped orders are distinct latest-revision order IDs, not raw extract rows. Returns are assigned to the original shipment month and translated at that month's ECB average quote. [S1–S3, S8]

Market	Orders	Units	Net sales	Contribution	Margin	Returned units
DEU	209	12,153	€625.8k	€262.4k	41.9%	73
FRA	209	11,880	€632.2k	€276.3k	43.7%	144
NLD	209	12,576	€697.7k	€315.6k	45.2%	144
POL	209	13,163	€761.9k	€356.5k	46.8%	180
CZE	209	13,733	€815.3k	€397.3k	48.7%	288
ESP	209	13,931	€850.8k	€421.9k	49.6%	252

The six markets produce €4,464,622 gross sales, €80,831 refunds, €4,383,792 net sales, €2,246,919 net COGS, €106,862 fulfillment and €2,030,010 contribution. Each component is rounded per order half-up before summing. Country totals equal the twelve monthly rows in the workbook and JSON. [S1–S3]

Data-quality reconciliation

Checkpoint	Treatment / result
Orders: 1,328 raw rows	1,297 IDs after 13 exact duplicate extras and 18 superseded revisions; no conflicting top revisions.
Eligibility	24 cancelled, 18 tests, one January 2026 shipment excluded; 1,254 eligible 2025 shipments. Reasons can overlap: cancelled rows also lack a 2025 shipment date.
Zero-price	12 eligible zero-price shipments retained; they carry units, COGS and fulfillment even if sales are zero.
Returns: 264 raw rows	243 IDs after 20 duplicate extras and one superseded revision; 241 linked by cutoff, one orphan quarantined, one post-cutoff excluded.
Missingness	No missing cost or FX inputs for eligible orders; no missing 2022–2024 World Bank values in the archived response.

Accounting interpretation

Booked net revenue is gross shipped sales less credited refunds, excluding taxes. Cash differs because payments, settlements, refund timing and actual bank FX are absent. Contribution then subtracts net product cost (restocked units recover cost at the original SKU rate) and nonrefundable fulfillment; it excludes hub capex, recurring hub fixed costs and wider corporate overhead. A returned unit is not automatically reusable. [S1–S3]

Monthly records, an order-level audit, return quarantine and exact formulas are in the accompanying workbook, JSON and analysis scripts.

2 | Market and currency context

The archived World Bank responses cover each market in 2022, 2023 and 2024. The table compares population change and 2024 GDP per capita in current US dollars; the latter is neither PPP nor real-income growth. These figures establish operating context, not product demand. [S6–S7]

Market	Population 2022	Population 2024	Change	GDP/head 2024 (US\$)
DEU	83,177,813	83,516,593	0.4%	56,104
FRA	68,184,457	68,551,653	0.5%	46,103
NLD	17,700,982	17,993,485	1.7%	67,465
POL	36,821,749	36,559,233	-0.7%	25,104
CZE	10,672,118	10,905,028	2.2%	31,828
ESP	47,786,102	48,848,840	2.2%	35,327

Spain (+2.22%) and Czechia (+2.18%) show the largest 2022–2024 population increases; Poland declined 0.71%. The Netherlands had the highest 2024 GDP/head in this set. None of these aggregate measures justifies the modeled 25% hub-related volume uplift; that is a client scenario assumption to validate. [S5–S7]

2025 ECB monthly reference-rate means

Local units per €1; mean of available published business-day observations. They are used for each order's sale month, including refunds credited later. EUR markets use 1.000000. [S1, S8]

Month	PLN/€	CZK/€	Month	PLN/€	CZK/€
2025-01	4.24665	25.16332	2025-07	4.25415	24.62461
2025-02	4.17223	25.07665	2025-08	4.26130	24.51662
2025-03	4.18201	25.00124	2025-09	4.25889	24.34677
2025-04	4.26522	25.03895	2025-10	4.24879	24.31539
2025-05	4.25380	24.92286	2025-11	4.23764	24.23405
2025-06	4.26577	24.80429	2025-12	4.22386	24.25929

The original source-room archive was retrieved 27 September 2026 and reports World Bank last-updated 13 July 2026. It can reflect revisions to 2022–2024 history; no live replacement series was blended into the calculations. [S6–S8]

3 | Hub choice and sensitivity

Year-zero capex is separate from recurring annual fixed cost. The scenario-policy formula applies the assumed 10% / 25% / 40% volume uplifts to 2025 contribution and per-unit savings, then subtracts annual hub fixed cost. The stress layers extra refunds of 3% of gross sales and, in PLN/CZK markets, depreciation of 10% of net sales into the base-uplift case. These are synthetic assumptions, not probabilities or causal effects. [S4–S5]

Feasible choice	Capex	FTE	Low annual	Base annual	High annual	Stress annual	Base payback
CZE+ESP	€225.0k	5	€64.6k	€198.1k	€331.6k	€32.2k	1.14y
POL+ESP	€240.0k	6	€58.6k	€186.3k	€314.1k	€29.3k	1.29y
POL+CZE	€205.0k	5	€56.9k	€179.0k	€301.1k	€-78.6k	1.15y
NLD+ESP	€270.0k	6	€44.0k	€166.9k	€289.9k	€107.6k	1.62y
NLD+CZE	€235.0k	5	€42.3k	€159.6k	€276.9k	€-0.2k	1.47y
ESP	€130.0k	3	€33.2k	€102.7k	€172.3k	€70.0k	1.27y
CZE	€95.0k	2	€31.5k	€95.4k	€159.3k	€-37.8k	1.00y
DEFER	€0.0k	0	€0.0k	€0.0k	€0.0k	€0.0k	—

The full 18-choice × 4-scenario feasible matrix, including all single hubs and pairs, is in the workbook and JSON. Germany + France breaches both caps; Germany + Spain and Germany + Poland breach the staffing cap. No pair synergy is assumed. [S4–S5]

Why this choice, and when it changes

Czechia + Spain leads the base case by €11,770 versus Poland + Spain and €31,168 versus Netherlands + Spain, while using €45,000 less capex and one fewer FTE than the latter. It also gives up €75,450 of annual contribution in the joint stress relative to Netherlands + Spain. Under the specified full 3% extra refund shock, a Czechia FX depreciation impact around 2.6% of its net sales would erase the base-case advantage over Netherlands + Spain; the policy stress assumes 10%. That switching point is arithmetic, not a forecast. [S4–S5]

Defer is financially preferable only if the assumed gains fail enough to make annual increments negative or if the service experiment does not validate a viable operating design. The board has not supplied a stress-risk tolerance or hurdle rate; the comparison therefore remains conditional on that preference. [S5]

All paybacks are simple capex / positive annual increment, with no ramp delay, taxes, discounting, working capital or residual value. They are not investment NPVs.

4 | Ninety-day execution and decision gates

Window	Accountable owner	Work and dependency	Exit gate
Days 0–30	COO / Finance controller	Freeze 2025 baseline and return definitions; contract carrier and facility options for Spain; map tax, labor and inventory controls before commitments.	Signed baseline; facility and carrier quotes within €130k Spain capex and three FTE plan.
Days 31–60	Spain operations lead / Customer service	Pilot Spain order routing and replenishment with limited SKUs; instrument promised versus actual delivery, availability, returns and true savings per unit.	At least 95% on-time delivery, ≥98% stock availability, and positive measured unit savings after variable handling.
Days 61–90	COO / CFO / Czechia lead	Review Spain pilot, Czechia customer-order evidence and PLN/CZK exposure; test current return mix, supplier quotes, staff readiness and revised payback.	Release Czechia ≤€95k and two FTE only if revised base increment remains positive and board accepts modeled stress exposure; otherwise retain Spain and reassess Netherlands.

The 95% and 98% pilot thresholds are proposed operational gates, not historical client performance. A two-site opening inside 90 days is not assumed; the day-90 decision releases or withholds second-site capex. [S4–S5]

KPI plan and ownership

KPI	Definition / cadence	Owner	Decision use
Incremental contribution	Pilot net sales – net COGS – fulfillment – local recurring cost, versus matched central-routing baseline; weekly and month-end.	Finance controller	Recalculate annualized payback and stop loss-making expansion.
Service level	On-time delivered orders / promised orders and in-stock order lines / requested lines; weekly.	Spain operations lead	Verify hub service advantage and gate operating readiness.
Returns	Returned physical units / shipped units; refund EUR / gross sales EUR by original sale cohort; weekly cohort readout.	Customer service lead	Detect refund shock and avoid treating returned units as full COGS recovery.
FX and capacity	PLN/CZK quote, settled cash rate, unit throughput per FTE and facility utilization; weekly.	Treasury / COO	Update Czechia stress and staffing capacity before commitment.

Principal risks

Uplift error: no causal estimate or control group in the extracts; use the Spain routing pilot and cohort comparison. FX/return shock: Czechia's stress annual increment is negative on its own (–€37,827); treasury and customer service must refresh the stress before release. Execution lag: permitting, supplier contracts and staffing can delay savings; CFO should revise the payback clock at each gate. Inventory tied up: working-capital needs are not in the model; finance must price them before final capital release. [S1–S5]

Source and calculation notes

[S1] Synthetic data dictionary, orders parts 1–2 and corrections; [S2] synthetic returns; [S3] synthetic effective-dated unit costs; [S4] synthetic hub-options.csv; [S5] synthetic scenario-policy.md; [S6] archived World Bank population response, SP.POP.TOTL; [S7] archived World Bank GDP/head response, NY.GDP.PCAP.CD; [S8] archived ECB eurofxref-hist.zip. Exact URLs, collection timestamps, SHA-256 hashes, periods and units are in sources/source_register.json. Scripts and quality exceptions are supplied with this brief. Source-room collection was local; public observations came from its archived official snapshots.

World Bank:

https://api.worldbank.org/v2/country/DEU;FRA;NLD;POL;CZE;ESP/indicator/SP.POP.TOTL?date=2022:2024&format=json&per_page=1000 and /NY.GDP.PCAP.CD with the same parameters. ECB: <https://www.ecb.europa.eu/stats/eurofxref/eurofxref-hist.zip>